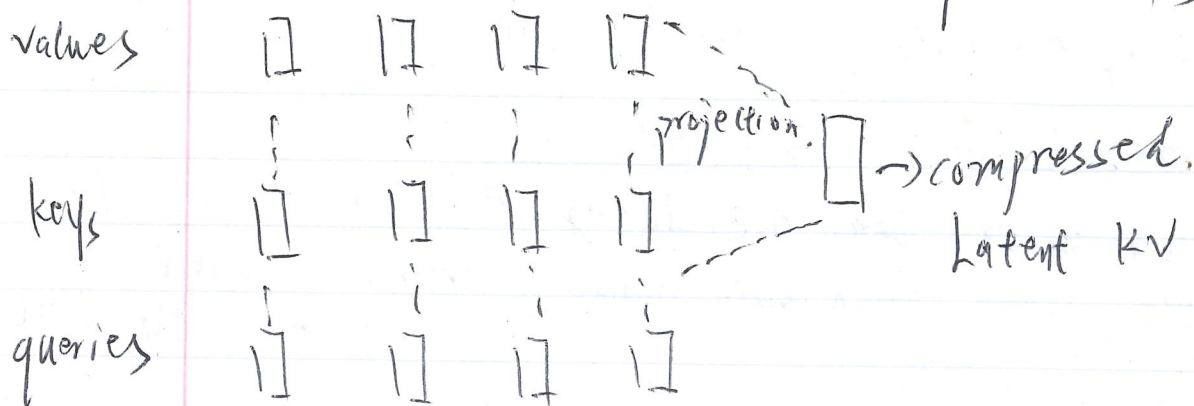


MLA: Multi-head Latent Attention (MLA)

↳ deepseek v_2/v_3



key idea: project down each key/value vector from $N \times H$ dim to C dim.

deepseek v_2 : reduce $N \times H = 16384$ to $C = 512$

↳ reduce $N \times H = 16384$ MLA not compatible with RoPE, so dims: 64 $\Rightarrow$ 512+64

latency / throughput improvements follow similarly from kv-cache

reduction as GQA

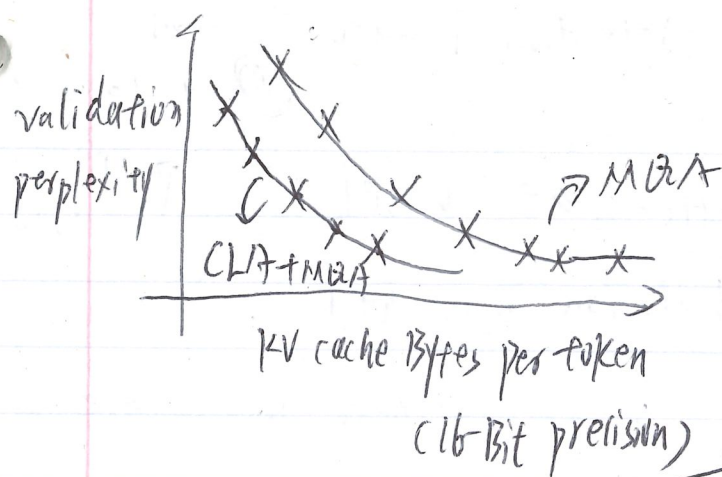
in Total: MHA ~~but~~ is more accurate than GQA
but MLA is simpler & better than MHA
and MLA is cheaper, lower memory cost
higher throughput
lower latency

Cross-layer attention (CLA)

↳ project K-V vectors from one attention layer to another attention layer in transformer architecture

$\Rightarrow$ share KVs across layers (just as GQA share KV across heads)

improves the pareto frontier of accuracy and KV cache size (latency/throughput)



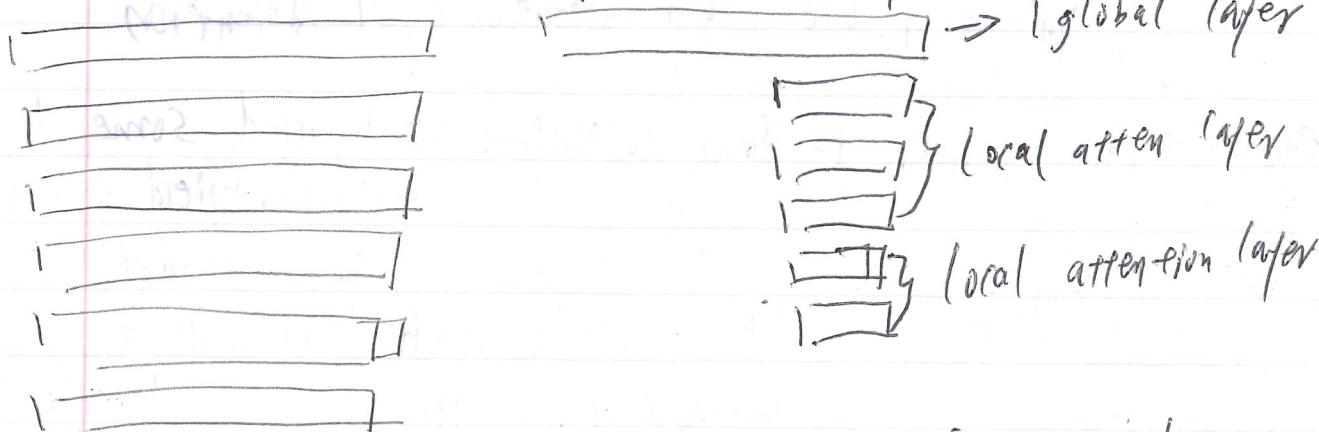
local attention: just look at the local context, which is the most relevant for modeling

Effective context scales linearly with the number of layers
KV cache is independent of sequence length.

problem: still hurts accuracy

solution: interleave local attention with global attention (Hybrid layers)

character.AI use 1 global layer every 6 layers



Summary: reduce KV cache size for memory-bound inference.

⇒ ① lower-dim KV cache (GQA, MLA, shared KV cache)

⇒ ② local attention on some layers